

Nikhil Sai Ramini

740-707-3866 | ramini.nikhilsai@gmail.com | [LinkedIn](#) | [GitHub](#) | [Portfolio](#)

SUMMARY

Software Engineer with a strong foundation in Python, Java, C, and C++ and over 2+ years of hands-on experience transitioning from Data Science to backend-focused development. Proven track record of building and optimizing scalable data pipelines, cloud-integrated systems, and high-performance backend services. Successfully developed and deployed efficient IoT solutions with a 30% increase in data transmission efficiency. Experienced in CI/CD pipelines, RESTful APIs, and containerized deployments using AWS and Docker. Passionate about designing clean, maintainable code and delivering robust, AI-driven software products in fast-paced environments.

EDUCATION

Ohio University

Aug 2023 - Dec 2024

Master of Science, Information & Telecommunication Systems

Athens, OH

- **GPA:** 4.0

- **Achievements:** Designed and deployed a LoRaWAN-based IoT sensor network on Ohio University's Green Roof, enabling real-time environmental monitoring and delivering data-driven insights for climate research and sustainability. Optimized the system to achieve a 30% increase in data transmission efficiency, enhancing overall performance and reliability.

Avanthi Scientific And Technological Research Academy

Jul 2019 - May 2023

Bachelor of Technology, Computer Science and Engineering

Hyderabad, India

- **GPA:** 3.6

- **Achievements:** Organized and participated in peer-led coding sessions and algorithm practice groups, mentoring juniors on introductory programming and helping classmates strengthen their Python, Java, and C skills through mock interviews and coding challenges. Elected Best Student President of Avanthi's Scientific and Technological Research Academy (2022–23), demonstrating exceptional leadership, organizational skills, and a commitment to fostering innovation and collaborative research among students.

WORK EXPERIENCE

Alpha Recon

Feb 2025 - Present

Data Scientist

- Developed and deployed scalable machine learning models using Python to address real-world business challenges and enhance user experience.
- Designed and maintained efficient data pipelines for automated data processing, model training, and inference, leveraging SQL databases where applicable.
- Generated comprehensive analytical reports and dashboards to support data-driven decision-making across teams.
- Collaborated with cross-functional teams including engineers and product managers to align technical solutions with business needs.
- Ensured data quality and integrity through systematic audits and implemented corrective actions.
- Authored detailed technical documentation and proof of concepts to facilitate reproducibility and stakeholder communication.

Ohio University

Apr 2024 - Dec 2024

IOT- Development, Programming, and Deployment/ Graduate Assistant

Athens, Ohio, US

- Developed, programmed, and deployed LoRaWAN IoT systems, enhancing communication efficiency and supporting integrated system operations.
- Configured and managed LoRaWAN systems by integrating Grove devices, Raspberry Pi with RAK HAT, sensors, gateways, and GPS modules for optimized IoT communication.
- Utilized Python for system configuration and data processing along with Arduino (C++) for microcontroller programming involving sensor integration, serial communication, and module configuration.
- Optimized data transmission efficiency by 30%, significantly improving overall system performance and reliability.
- Conducted comprehensive functional, performance, power, and stress testing to resolve hardware-software issues and enhance system robustness.
- Deployed a robust LoRaWAN system capable of handling real-time communication and data exchange between sensors and servers while focusing on user experience.

Ohio university

Aug 2023 - Mar 2024

Technical Support/ Graduate Assistant

Athens, Ohio, US

- Provided technical troubleshooting and maintenance for lab computers, printers, and network systems, ensuring optimal functionality and minimal downtime.
- Assisted students, faculty, and staff with resolving technical issues, delivering hands-on support and user training as needed.
- Resolved tickets in the OIT department, addressing a range of technical issues and ensuring timely solutions.
- Configured and deployed new workstations, peripherals, and network setups, ensuring seamless integration and operation.
- Documented technical issues, resolutions, and procedures to create a knowledge base and improve support efficiency.
- Streamlined lab operations by implementing system updates, monitoring resource usage, and ensuring compliance with institutional policies.

PROJECTS

BARN Zero Trust Model

- Designed and implemented a Zero Trust network architecture based on the BARN model to enhance organizational security. Utilized principles of continuous verification, least privilege access, and micro-segmentation to mitigate internal and external threats.

IPv6 Deployment for Ohio University

- Led the planning and deployment of IPv6 within the university network in Infoblox. Conducted address planning, DNS configuration, and router interface setups to ensure seamless coexistence with IPv4 and future-proof network infrastructure.

ML Techniques for Cyber Attacks Detection

- Developed a cyber attack detection system using machine learning algorithms such as decision trees and neural networks. Implemented the solution using Python, scikit-learn, and TensorFlow to classify network anomalies with high accuracy.

Stock Market Analysis and Prediction

- Built a predictive model for stock price movement using historical market data and technical indicators. Employed Python with libraries like pandas, scikit-learn, and matplotlib for data analysis, feature engineering, and model evaluation.

COP FLIX - Cinematic Virtual Reality Streaming Service

- Created a VR-based streaming platform offering immersive cinematic experiences. Engineered the system with Unity and C#, integrating 360-degree video, spatial audio, and user interactivity for an engaging virtual entertainment solution.

SKILLS

- **Programming & Development:** Python, C, Java, C++, R, JavaScript, HTML, CSS, Powershell, User Experience Focus, React, Objective-C, Data Science, Object Oriented Programming
- **Data & Cloud Engineering:** SQL, MYSQL, Data Engineering, MongoDB, Data Pipelines, AWS, Cloud Infrastructure, CI/CD Workflows, Azure, Docker, Cloud Computing
- **Artificial Intelligence:** AI, Machine Learning, RLHF, Generative AI, Prompt Engineer
- **Systems & Tools:** VoIP, IoT Systems Development, Programming, Deployment, DHCP, DNS, IPv4, IPv6, SDN, Infoblox, 5G Wireless, Distributed Systems, Code Review, Software Integration, Technical Documentation, Troubleshooting, Microcontrollers, Tableau, Rest APIs, Kubernetes, Linux Kernel, Backend Development, Docker

CERTIFICATES & BADGES

- 5G Wireless Certification by Telecommunications Education Center (TEC)
- Python Essentials 1&2 by Cisco
- Junior Cybersecurity Analyst by Cisco
- Tata Group's Cybersecurity Analyst Job Stimulation
- Python Certification by Kaggle
- Introduction to Cybersecurity by Cisco

ACHIEVEMENTS

- Earned a Gold Badge in Python on HackerRank, showcasing advanced problem-solving skills and expertise in Python programming
- Achieved Silver Badges in both C and Java on HackerRank, demonstrating proficiency in multiple programming languages and strong coding abilities.
- Designed and deployed a LoRaWAN-based IoT sensor network on Ohio University's Green Roof, enabling real-time environmental monitoring and providing data-driven insights to support climate research and sustainability initiatives.
- Best Student President Avanthi's Scientific And Technological Research Academy 2022-23